YUHAO BAO

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RESEARCH INTERESTS

HCI and AI: how model outputs compare with human labels, how they change across model versions and repeated runs, and when people can rely on them in crisis response.

EDUCATION

Brigham Young University, Provo, UT — Expected April 2027
B.S. in Computer Science; Minors in Mathematics and Physics. GPA: 3.85/4.00 (Computer Science: 3.90/4.00)

PUBLICATIONS AND MANUSCRIPTS

Bao, Y., & Hughes, A. L. (2026). When Social Media Images Need Words: Measuring Context Gap and Fusion Tax in Crisis Image Captioning. *Proceedings of the International ISCRAM Conference, 23*. Work-in-Progress paper. doi:10.59297/4bar0c32

Bao, Y., & Hughes, A. L. Same Tweets, Different Labels: Version Updates and Rerun Variation in LLM Urgency Labeling of Hurricane Dorian Tweets (working title). In preparation for the International ISCRAM Conference 2027.

Liang, H., et al. (third author: **Bao, Y.**). A user study of avatar appearance and locomotion in virtual reality. Manuscript in preparation.

RESEARCH EXPERIENCE

Research Assistant, HCI Research Group, BYU (Advisor: Prof. Amanda Hughes) — Nov 2023 – Present

LLM urgency labeling of Hurricane Dorian tweets (first-author manuscript, in preparation) — Feb 2026 – Present

- Compared six versions of Qwen, Gemini, and GPT with historical TREC-IS labels on 367 Hurricane Dorian tweets with 449 photos, labeling urgency (Priority) and information type, with and without the photos, three runs per condition.
- Agreement with the historical labels hid the direction of change: in all three families, the newer version raised more urgency labels than it lowered.
- Deployed Qwen on BYU's GPU cluster. Repeated runs of the same version also changed labels, most for Qwen; the two Claude versions are still to be run.

Crisis image captioning with vision-language models (ISCRAM 2026) — Jun 2025 – Jan 2026

- Compared Gemini 2.0 Flash, Qwen2.5-VL, and BLIP on 204 crisis image-post pairs with and without post text; adding text raised Gemini's accuracy from 75.5% to 91.7% but also introduced text-driven errors.
- Designed the three-level scoring rubric and six-category error taxonomy and served as primary coder for 1,020 scorable captions; a second coder's independent subset of 100 gave Krippendorff's $\alpha = 0.69$.

Collaborative learning of ChatGPT on social media (lab manuscript, working draft) — Nov 2023 – Jun 2025

- Wrote analysis code and produced figures on trends, word frequency, and sentiment for popular and random samples drawn from a corpus of about 5.4 million tweets discussing ChatGPT.
- Drafted much of the Results and parts of the Methods (data collection, coding process, VADER sentiment analysis).

Text-to-SQL with Retrieval and LoRA, BYU (Advisor: Prof. Porter Jenkins) — Sep 2025 – Apr 2026

- Fine-tuned Qwen3.5-9B with LoRA in a graduate student's retrieval-augmented Text-to-SQL project on a real commercial database.
- Implemented Shadow-LoRA (Base-to-Instruct adapter transfer) against a matched Direct-LoRA baseline; exact match on 1,089 validation examples rose from 19.47% to 30.49%.

VR avatar and locomotion study, Advanced 3D Computer Graphics Lab (Prof. Parris Egbert) — May 2026 – Sep 2026

- With first author Hunter Liang, analyzed data from a 20-participant, 2 × 3 within-subject VR study of how avatar appearance and locomotion relate to task performance and embodiment; tested avatar-by-locomotion interactions with mixed-effects models, corrected for multiple comparisons, and contributed to the Statistical Analysis, Results, and Discussion sections.

SKILLS

Research methods: rubric and error-taxonomy design, qualitative coding, inter-rater reliability, mixed-effects models
Programming: Python, SQL, Java, C++, JavaScript/TypeScript, Git, Linux
ML and data: PyTorch, Hugging Face Transformers, LoRA/PEFT, vLLM, Slurm, pandas, NumPy, PostgreSQL

PRESENTATIONS AND CONFERENCES

Bridging the Context Gap in Crisis Image Captioning. Oral presentation, BYU Student Research Conference, Feb 2026.

Collaborative Learning of AI Tools: Insights from ChatGPT Discussions on Social Media. Oral presentation, BYU Student Research Conference, Mar 2025.

ACM CSCW 2026, Salt Lake City: attended with BYU Computer Science travel funding, Oct 2026.